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SEMICONDUCTOR DEVICE AND POWER CONVERSION DEVICE

Abstract

There is provided a semiconductor device in which the occurrence of a short-circuit failure is suppressed. A semiconductor device includes: a substrate; a semiconductor element; a terminal; and a metal wire. The semiconductor element is disposed on the substrate. The terminal is disposed at a position farther than the semiconductor element when viewed from the substrate. The metal wire connects the semiconductor element and the terminal. A notch portion is provided in the terminal. A part of the metal wire is inserted into the notch portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This nonprovisional application is based on Japanese Patent Application No. 2024-033678 filed on Mar. 6, 2024 with the Japan Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a semiconductor device and a power conversion device.

Description of the Background Art

[0003] A wire bonding method using a metal wire has been conventionally known as a method for forming internal wiring in a semiconductor device (refer to, for example, Japanese Utility Model Laying-Open No. 1-163345 (full text)). In Japanese Utility Model Laying-Open No. 1-163345, a metal wire is formed along a direction in which a terminal extends.

SUMMARY OF THE INVENTION

[0004] However, in a semiconductor device of transfer mold type, a metal wire may be deformed by a mold resin when the mold resin is injected as a sealing resin. The deformation of the metal wire may consequently cause a short-circuit failure.

[0005] The present disclosure has been made to solve the above-described problem, and an object of the present disclosure is to provide a semiconductor device in which the occurrence of a short-circuit failure is suppressed.

[0006] A semiconductor device according to the present disclosure includes: a substrate; a semiconductor element; a terminal; and a metal wire. The semiconductor element is disposed on the substrate. The terminal is disposed at a position farther than the semiconductor element when viewed from the substrate. The metal wire connects the semiconductor element and the terminal. A notch portion is provided in the terminal. A part of the metal wire is inserted into the notch portion.

[0007] A power conversion device according to the present disclosure includes: a main conversion circuit; and a control circuit. The main conversion circuit has the above-described semiconductor device, and converts input power and outputs the converted input power. The control circuit outputs a control signal for controlling the main conversion circuit to the main conversion circuit.

[0008] The foregoing and other objects, features, aspects and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a side view of a semiconductor device according to a first embodiment.

[0010] FIG. **2** is a plan view of the semiconductor device according to the first embodiment.

[0011] FIG. **3** is a partially enlarged side view of a region III shown in FIG. **1**.

[0012] FIG. **4** is a plan view of a signal terminal according to the first embodiment.

[0013] FIG. **5** is a side view of a semiconductor device according to a second embodiment.

[0014] FIG. **6** is a plan view of the semiconductor device according to the second embodiment.

[0015] FIG. **7** is a side view of a semiconductor device according to a third embodiment.
[0016] FIG. **8** is a plan view of the semiconductor device according to the third embodiment.
[0017] FIG. **9** is a plan view of a signal terminal according to the third embodiment.
[0018] FIG. **10** is a side view of a semiconductor device according to a fourth embodiment.
[0019] FIG. **11** is a partially enlarged side view of a region XI shown in FIG. **10**.
[0020] FIG. **12** is a plan view of a signal terminal according to a fifth embodiment.
[0021] FIG. **13** is a plan view of a signal terminal according to a sixth embodiment.
[0022] FIG. **14** is a block diagram showing a configuration of a power conversion system to which a power conversion device according to a seventh embodiment is applied.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0023] Hereinafter, embodiments of the present disclosure will be described. Unless otherwise mentioned, the same or corresponding portions in the following drawings are denoted by the same reference numerals, and description thereof will not be repeated.

First Embodiment

<Configuration of Semiconductor Device>

[0024] FIG. **1** is a side view of a semiconductor device **100** according to a first embodiment. FIG. **2** is a plan view of semiconductor device **100** according to the first embodiment. FIG. **3** is a partially enlarged side view of a region III shown in FIG. **1**. FIG. **4** is a plan view of a signal terminal **41** according to the first embodiment.
[0025] Semiconductor device **100** shown in FIGS. **1** to **4** is, for example, power semiconductor device **100** and mainly includes a substrate **1**, a semiconductor element **2**, a joining portion **3**, a terminal **4**, a metal wire **6**, and a sealing portion **7**. In FIGS. **1** and **2**, sealing portion **7** is indicated by a dotted line.
[0026] Substrate **1** has a heat spreader **11**, an insulating sheet **12** and a metal foil **13**. Heat spreader **11** is disposed on insulating sheet **12**. Insulating sheet **12** is disposed on metal foil **13**. Metal foil **13** is disposed on a surface of insulating sheet **12** located opposite to a surface on which heat spreader **11** is disposed.
[0027] A material of heat spreader **11** is, for example, copper (Cu). As shown in FIG. **2**, heat spreader **11** includes a plurality of heat spreader portions **11***a* and **11***b*. The plurality of heat spreader portions **11***a* and **11***b* are disposed to be spaced apart from each other in the y direction.
[0028] As shown in FIG. **2**, substrate **1** has a main surface **11***s*. Main surface **11***s* is a surface to which semiconductor element **2** is electrically connected. Heat spreader portion **11***a* has a surface **11***sa*. Heat spreader portion **11***b* has a surface **11***sb*. Surfaces **11***sa* and **11***sb* form main surface **11***s* of substrate **1**.
[0029] As shown in FIGS. **1** and **2**, a direction perpendicular to main surface **11***s* is defined as the z direction. The x direction and the y direction are directions perpendicular to the z direction. The y direction is a direction perpendicular to the x direction. That is, main surface **11***s* is a surface extending in the x direction and the y direction.
[0030] Insulating sheet **12** is connected to a surface of heat spreader **11** located opposite to surface **11***sa* and surface **11***sb*. Insulating sheet **12** may be, for example, an insulating sheet containing inorganic powder or glass fiber. Metal foil **13** and heat spreader **11** are electrically insulated by insulating sheet **12**.
[0031] A material of metal foil **13** may be a metal having high thermal conductivity. The material of metal foil **13** may be, for example, any material selected from the group consisting of aluminum, copper, iron, and nickel, or may be an alloy containing at least any material selected from the group consisting of aluminum, copper, iron, and nickel.
[0032] As shown in FIGS. **1** and **2**, in a plan view of main surface **11***s*, heat spreader **11** may be smaller than each of insulating sheet **12** and metal foil **13** in the x direction and the y direction.
[0033] Substrate **1** having a three-layer structure constituted by heat spreader **11**, insulating sheet **12** and metal foil **13** may be used as substrate **1**, or an insulating substrate may be used as substrate

1. The insulating substrate may be constituted by a base plate, an insulating layer and a circuit pattern. The insulating layer may be disposed on the base plate, and the circuit pattern may be disposed on the insulating layer. Each of the base plate and the circuit pattern is made of metal such as copper, for example. The insulating layer ensures electrical insulation from the outside of semiconductor device **100**. A material of the insulating layer may be, for example, inorganic ceramic, or may be a material having ceramic powder dispersed in a thermosetting resin such as an epoxy resin.

[0034] Semiconductor element **2** includes a plurality of semiconductor element portions **21**, **22**, **23**, and **24**. As shown in FIG. **1**, semiconductor element **2** is connected to main surface **11***s* of substrate **1** with joining portion **3** interposed therebetween. Specifically, each of semiconductor element portions **21** and **22** is connected to surface **11***sa* with joining portion **3** interposed therebetween. Each of semiconductor element portions **23** and **24** is connected to surface **11***sb* with joining portion **3** (not shown) interposed therebetween.

[0035] A material of joining portion **3** may be, for example, solder, or may be an electrically conductive adhesive, or may be a joining material containing silver (Ag) particles or copper (Cu) particles having sinterability. By using the joining material having sinterability to join semiconductor element **2** and substrate **1**, the heat dissipation properties and lifespan of joining portion **3** are improved, as compared with when solder is used.

[0036] Semiconductor element **2** is so-called power semiconductor element **2** that controls electric power. A material of semiconductor element **2** may be silicon (Si), or may be silicon carbide (SiC), gallium nitride (GaN), gallium oxide (Ga**203**), or diamond as a wide bandgap semiconductor material. When such a so-called wide bandgap semiconductor material having a wider bandgap than that of silicon is used as the material of semiconductor element **2**, semiconductor device **100** with high efficiency and compatible with high temperatures can be obtained.

[0037] Semiconductor element **2** may be of any kind. For example, an insulated gate bipolar transistor (IGBT), a free wheel diode (FWD), or a metal oxide semiconductor field effect transistor (MOSFET) can be used.

[0038] As shown in FIG. **2**, semiconductor element portions **21** and **22** are disposed to be spaced apart from each other in the x direction. Semiconductor element portions **23** and **24** are disposed to be spaced apart from each other in the x direction.

[0039] As shown in FIG. **2**, each of semiconductor element portions **21** and **23** has a main electrode **81** and three signal electrodes **82**. Main electrode **81** and three signal electrodes **82** are provided on a surface (upper surface) of each of semiconductor element portions **21** and **23**. Each of semiconductor element portions **22** and **24** has a main electrode **83**. Main electrode **83** is provided on a surface (upper surface) of each of semiconductor element portions **22** and **24**. Main electrode **81** is disposed to be spaced apart from each of three signal electrodes **82** in the x direction. Three signal electrodes **82** may be disposed to be spaced apart from each other at equal intervals in the y direction.

[0040] Terminal **4** has a main terminal **42** and a plurality of signal terminals **41**. As shown in FIG. **2**, main terminal **42** includes a plurality of main terminal portions **42***a*, **42***b* and **42***c*. Metal wire **6** connects semiconductor element **2** and terminal **4**. Metal wire **6** includes a plurality of metal wire portions **61**, **62**, **63**, and **64**. In semiconductor device **100** according to the first embodiment, six metal wire portions **61**, six metal wire portions **62**, three metal wire portions **63**, and one metal wire portion **64** are provided.

[0041] Each of three metal wire portions **62** connects main terminal portion **42***a*, main electrode **83** of semiconductor element portion **22**, and main electrode **81** of semiconductor element portion **21** by a wire bonding method. Each of three metal wire portions **62** connects main terminal portion **42***c*, main electrode **83** of semiconductor element portion **24**, and main electrode **81** of semiconductor element portion **23** by a wire bonding method. In this way, main terminal **42** through which a main current flows is electrically connected to main electrodes **81** and **83** that

input and output main power.
[0042] Each of six metal wire portions **61** connects each of six signal terminals **41** and each of six signal electrodes **82** of semiconductor element portions **21** and **23** by a wire bonding method. In this way, signal terminal **41** through which a control signal flows is electrically connected to signal electrode **82** that inputs and outputs a control signal and a sense signal.
[0043] As shown in FIG. **2**, each of three metal wire portions **63** may connect main electrode **81** of semiconductor element portion **21** and surface **11***sb* of heat spreader portion **11***b* by a wire bonding method. Metal wire portion **64** may connect main terminal portion **42***b* and surface **11***sa* of heat spreader portion **11***a* by a wire bonding method.
[0044] A material of metal wire **6** may be any metal, and may be, for example, aluminum (Al) or copper (Cu). A diameter of metal wire **6** may be, for example, equal to or more than 80 μm and equal to or less than 600 μm. Particularly, a diameter of metal wire portion **62** through which the main current flows may be 400 μm. On the other hand, a diameter of metal wire portion **61** through which the control signal and the like flow may be 200 μm.
[0045] A material of terminal **4** is, for example, copper (Cu). The material of terminal **4** may be any material having heat dissipation properties in addition to electrically conductive properties. The material of terminal **4** may be, for example, an alloy containing copper or aluminum, or may be a composite material obtained by stacking these metals.
[0046] A part of each of the plurality of main terminal portions **42***a*, **42***b* and **42***c* and the plurality of signal terminals **41** extends in the x direction from semiconductor element **2** to the outside of sealing portion **7**. Signal terminal **41** extends along a direction opposite to a direction in which main terminal **42** extends.
[0047] Sealing portion **7** covers substrate **1**, semiconductor element **2**, joining portion **3**, a part of main terminal **42**, a part of signal terminal **41**, and metal wire **6**. As shown in FIG. **1**, a surface of metal foil **13** located opposite to a surface connected to insulating sheet **12** may be exposed from sealing portion **7**. When semiconductor device **100** is incorporated into a power conversion device or the like, metal foil **13** may be connected to a heat dissipation fin in the portion of metal foil **13** exposed from sealing portion **7**. In order to electrically insulate metal foil **13** from the outside, sealing portion **7** may cover the whole of metal foil **13**.
[0048] A part of terminal **4** extends from a surface of sealing portion **7** to the outside such that terminal **4** can be connected to an external device outside sealing portion **7**. The portion of terminal **4** extending to the outside of sealing portion **7** may be bent by, for example, forming. A conductor (not shown) such as a wire or a terminal for electrical connection to a circuit board or another semiconductor device may be connected to the above-described portion of terminal **4**. Any method can be used as a method for connecting the conductor and the above-described portion, and the conductor and the above-described portion may be fixed by, for example, a fixing member such as a screw.
[0049] A material of sealing portion **7** may be a resin having insulating properties. The resin having insulating properties may be, for example, an epoxy resin. Sealing portion **7** may be formed by transfer molding.
[0050] As described above, as metal wire **6**, metal wire portion **61** connects semiconductor element **2** and terminal **4**. When signal terminal **41** is disposed as terminal **4** at the same level as semiconductor element **2** in the z direction, a position in the z direction at which metal wire portion **61** is connected to signal terminal **41** is the same as a position at which metal wire portion **61** is connected to semiconductor element **2**.
[0051] The diameter of metal wire portion **61** is smaller than the diameter of metal wire portion **62**. Therefore, when a resin is injected during formation of sealing portion **7**, metal wire portion **61** may be crushed along the −z direction and deformed due to an injection pressure of the resin. When deformed metal wire portion **61** comes into contact with a member (e.g., heat spreader **11**) having a potential different from a potential of metal wire portion **61**, a short-circuit failure occurs in

semiconductor device **100**.

[0052] Furthermore, when the plurality of metal wire portions **61** are disposed to be adjacent to each other, metal wire portions **61** adjacent to each other may come into contact with each other during manufacturing of semiconductor device **100**, due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin. As a result, a short-circuit failure occurs in semiconductor device **100**.

[0053] Semiconductor device **100** according to the first embodiment is characterized in that a notch portion h is provided in terminal **4** as shown in FIG. **4**. By disposing terminal **4** such that a part of metal wire **6** is inserted into notch portion h, deformation and movement of metal wire **6** are limited by notch portion h. As a result, contact between metal wire **6** and the other members, and contact between metal wires **6** adjacent to each other are suppressed. In this way, the occurrence of a short-circuit failure in semiconductor device **100** is suppressed.

[0054] In addition, signal terminal **41** is disposed at a position farther than semiconductor element **2** when viewed from substrate **1**. Therefore, metal wire **6** on the connection portion **6***a* side connected to semiconductor element **2** with respect to a center of metal wire **6** in the x direction is held, and thus, deformation of metal wire **6** is further suppressed.

[0055] As shown in FIGS. **3** and **4**, signal terminal **41** has an upper surface **41***c*, a lower surface **41***d*, a tip end surface **41***s***1**, and a pair of side surfaces **41***s***2**. Upper surface **41***c* and lower surface **41***d* are surfaces substantially perpendicular to the z direction. Upper surface **41***c* and lower surface **41***d* face each other in the z direction. Notch portion h passes through signal terminal **41** to extend from upper surface **41***c* to lower surface **41***d*.

[0056] As shown in FIG. **4**, signal terminal **41** extends in the x direction within sealing portion **7**. Each of tip end surface **41***s***1** and the pair of side surfaces **41***s***2** is continuous to upper surface **41***c* and lower surface **41***d*. Tip end surface **41***s***1** may be a surface substantially perpendicular to the x direction. The pair of side surfaces **41***s***2** face each other in the y direction. The pair of side surfaces **41***s***2** may be surfaces substantially perpendicular to the y direction. Notch portion h may be opened to tip end surface **41***s***1**. Notch portion h may be disposed at a position where notch portion h is sandwiched between the pair of side surfaces **41***s***2** in the y direction, or may be provided at a center of signal terminal **41** in the width direction (in the first embodiment, the y direction).

[0057] A distance t**2** from main surface **11***s* of substrate **1** to upper surface **41***c* of signal terminal **41** is longer than a distance t**1** from main surface **11***s* of substrate **1** to a surface of semiconductor element portion **21**. Thus, a part of metal wire portion **61** is inserted into notch portion h.

[0058] From a different point of view, as shown in FIG. **3**, metal wire portion **61** includes vertex portion **6**T and connection portions **6***a* and **6***b*. Vertex portion **6**T is a portion of metal wire portion **61** disposed at a position farthest from semiconductor element **2** in the z direction. Connection portion **6***a* is a portion of metal wire portion **61** that is in contact with signal electrode **82**. Connection portion **6***b* is a portion of metal wire portion **61** that is in contact with signal terminal **41**.

[0059] As described above, signal terminal **41** is disposed at the position farther than semiconductor element **2** when viewed from substrate **1**. Therefore, connection portion **6***b* is disposed at a position farther than connection portion **6***a* in the z direction when viewed from semiconductor element portion **21**. In addition, metal wire portion **61** is formed to be curved. Therefore, vertex portion **6**T is disposed at a position farther than connection portion **6***b* in the z direction when viewed from semiconductor element portion **21**. From a different point of view, a distance t**3** from substrate **1** to vertex portion **6**T is longer than distance t**1** from substrate **1** to connection portion **6***a*, and is longer than distance t**2** from substrate **1** to connection portion **6***b*. Thus, a tip end of signal terminal **41** can suppress deformation of metal wire portion **61** in the −z direction, and suppress contact with the other members in metal wire portion **61**.

[0060] A part of metal wire portion **61** is surrounded by notch portion h in a plan view of main surface **11***s*. Thus, contact between metal wires **6** adjacent to each other can be suppressed. Even if metal wire **6** comes into contact with an inner wall of terminal **4** that forms notch portion h, the characteristics of semiconductor device **100** are not affected because metal wire **6** and terminal **4** have an equal potential.

[0061] When a width w of notch portion h in the y direction is large, an amount of movement of metal wire portion **61** is large. When width w of notch portion h in the y direction is small, it is difficult to insert metal wire portion **61** into notch portion h. Therefore, width w of notch portion h in the y direction may be equal to or more than 1.5 times and equal to or less than 2 times the diameter of metal wire portion **61**.

[0062] Furthermore, as shown in FIG. **3**, a part of a region of metal wire portion **61** from connection portion **6***a* to vertex portion **6**T is inserted into notch portion h (see the dotted line in FIG. **3**). From a different point of view, notch portion h is disposed between vertex portion **6**T and connection portion **6***a* in the x direction. Particularly, notch portion h may be disposed on the connection portion **6***a* side with respect to a center of metal wire portion **61** in the x direction. Thus, since notch portion h holds metal wire portion **61** on the connection portion **6***a* side, deformation of metal wire portion **61** is further suppressed. As a result, the occurrence of a short-circuit failure in semiconductor device **100** is suppressed.

[0063] Notch portion h may only be disposed between vertex portion **6**T and connection portion **6***a* in the z direction. Notch portion h may be disposed at a center between vertex portion **6**T and connection portion **6***a* in the z direction. Notch portion h may be disposed on the vertex portion **6**T side when viewed from the center between vertex portion **6**T and connection portion **6***a*. Notch portion h may be disposed on the connection portion **6***a* side when viewed from the center between vertex portion **6**T and connection portion **6***a*.

[0064] When signal terminal **41** is disposed so as not to overlap with substrate **1**, metal wire portion **61** becomes longer. As a result, semiconductor device **100** increases in size in the x direction. In addition, when metal wire portion **61** becomes longer, deformation of metal wire portion **61** may become larger during manufacturing of semiconductor device **100**, due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin.

[0065] Therefore, as shown in FIGS. **1** to **3**, a part of signal terminal **41** may be disposed at a position where the part of signal terminal **41** overhangs substrate **1**. Specifically, a part of signal terminal **41** may overlap with substrate **1** in a plan view of main surface **11***s*. Thus, since signal terminal **41** is disposed such that a part thereof overlaps with substrate **1**, the length of metal wire portion **61** can be reduced even when substrate **1** is increased in size. As a result, an increase in size of semiconductor device **100** in the x direction can be suppressed.

[0066] Signal terminal **41** includes a tip end region **41***a* and an external region **41***b*. Tip end region **41***a* is a region where signal terminal **41** overlaps with substrate **1** in a plan view of main surface **11***s*. External region **41***b* is a region of signal terminal **41** other than tip end region **41***a*, and is a region where signal terminal **41** does not overlap with substrate **1** in a plan view of main surface **11***s*.

[0067] Notch portion h is provided in tip end region **41***a* of signal terminal **41**. Connection portion **6***b* is connected to external region **41***b* of signal terminal **41** in the x direction. That is, connection portion **6***b* is disposed at a position that does not overlap with substrate **1** in a plan view of main surface **11***s*.

[0068] Thus, an increase in size of semiconductor device **100** in the x direction can be suppressed. With an increase in size of substrate **1**, heat spreader **11** may be increased in size. As a result, the heat dissipation properties of semiconductor device **100** can be improved. In addition, by reducing the length of metal wire portion **61**, deformation of metal wire portion **61** during manufacturing of semiconductor device **100** due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin can be suppressed.

[0069] The example in which notch portion h is provided in signal terminal **41** has been described.

However, notch portion h may be provided in main terminal **42**, or the above-described configuration may be applied to main terminal **42** and metal wire portions **62** and **64**.

Functions and Effects

[0070] Semiconductor device **100** according to the present disclosure includes substrate **1**, semiconductor element **2**, terminal **4**, and metal wire **6**. Semiconductor element **2** is disposed on substrate **1**. Terminal **4** is disposed at the position farther than semiconductor element **2** when viewed from substrate **1**. Metal wire **6** connects semiconductor element **2** and terminal **4**. Notch portion h is provided in terminal **4**. A part of metal wire **6** is inserted into notch portion h.

[0071] With such a configuration, contact between metal wire **6** and the other members, and contact between metal wires **6** adjacent to each other are suppressed. As a result, the occurrence of a short-circuit failure in semiconductor device **100** is suppressed.

[0072] According to semiconductor device **100** described above, terminal **4** includes tip end region **41***a* and external region **41***b*. Tip end region **41***a* overlaps with substrate **1** in a plan view of substrate **1**. External region **41***b* is a region other than tip end region **41***a*. Metal wire **6** is connected to external region **41***b*.

[0073] With such a configuration, an increase in size of semiconductor device **100** in the direction in which terminal **4** extends can be suppressed. In addition, by reducing the length of metal wire **6**, deformation of metal wire **6** during manufacturing of semiconductor device **100** due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin can be suppressed.

[0074] According to semiconductor device **100** described above, notch portion h is provided at the center of terminal **4** in the width direction.

[0075] With such a configuration, contact between metal wire **6** and the other members, and contact between metal wires **6** adjacent to each other are suppressed. As a result, the occurrence of a short-circuit failure in semiconductor device **100** is suppressed.

Second Embodiment

<Configuration of Semiconductor Device>

[0076] FIG. **5** is a side view of semiconductor device **100** according to a second embodiment. FIG. **5** corresponds to FIG. **1**. FIG. **6** is a plan view of semiconductor device **100** according to the second embodiment. FIG. **6** corresponds to FIG. **2**. Semiconductor device **100** shown in FIGS. **5** and **6** basically has the same configuration as that of semiconductor device **100** shown in FIGS. **1** to **4** and can achieve the same effect as that of semiconductor device **100** shown in FIGS. **1** to **4**. However, semiconductor device **100** shown in FIGS. **5** and **6** is different from semiconductor device **100** shown in FIGS. **1** to **4** in that main electrodes **81** and **83** (see FIG. **2**) are connected to main terminal **42** with joining portions **3** interposed therebetween.

[0077] As shown in FIG. **6**, main terminal portion **42***a* may extend to a position that overlaps with main electrode **81** of semiconductor element portion **21** in the x direction. Main terminal portion **42***b* may have a protruding portion **44** extending in the x direction. Main terminal portion **42***c* may extend to a position that overlaps with main electrode **81** of semiconductor element portion **23** in the x direction.

[0078] As shown in FIG. **5**, main terminal portion **42***a* is connected to main electrode **81** of semiconductor element portion **21** with joining portion **3** interposed therebetween. Main terminal portion **42***a* is connected to main electrode **83** of semiconductor element portion **22** with joining portion **3** interposed therebetween. Main terminal portion **42***a* may have a protruding portion **43**. Protruding portion **43** extends from main terminal portion **42***a* in the y direction to overlap with surface **11***sb* of heat spreader portion **11***b*. Protruding portion **43** is connected to surface **11***sb* of heat spreader portion **11***b* with joining portion **3** (not shown) interposed therebetween.

[0079] Protruding portion **44** is connected to surface **11***sa* of heat spreader portion **11***a* with joining portion **3** (not shown) interposed therebetween.

[0080] As shown in FIG. **6**, main terminal portion **42***c* is connected to main electrode **81** of semiconductor element portion **23** with joining portion **3** (not shown) interposed therebetween. Main terminal portion **42***c* is connected to main electrode **83** of semiconductor element portion **24** with joining portion **3** (not shown) interposed therebetween.

[0081] By connecting main terminal **42** to main electrodes **81** and **83** with joining portions **3** (made of solder) interposed therebetween as described above, the heat cycle resistance and lifespan of semiconductor device **100** are improved, as compared with when main terminal **42** and main electrodes **81** and **83** are connected to each other with metal wire portions **62** interposed therebetween.

[0082] A material of joining portions **3** that connect main terminal **42** and main electrodes **81** and **83** may be, for example, solder, or may be an electrically conductive adhesive, or may be a joining material containing silver (Ag) particles or copper (Cu) particles having sinterability. By using the joining material having sinterability to join main terminal **42** and main electrodes **81** and **83**, the heat dissipation properties and lifespan of joining portions **3** are improved, as compared with when solder is used.

<Functions and Effects>

[0083] According to semiconductor device **100** described above, semiconductor element **2** has main electrode **81** and signal electrode **82**. Terminal **4** has main terminal **42** and signal terminal **41**. Main terminal **42** is connected to main electrode **81** with joining portion **3** interposed therebetween. Signal terminal **41** is connected to signal electrode **82** via metal wire **6**.

[0084] With such a configuration, the heat cycle resistance and lifespan of semiconductor device **100** are improved.

Third Embodiment

<Configuration of Semiconductor Device>

[0085] FIG. **7** is a side view of semiconductor device **100** according to a third embodiment. FIG. **7** corresponds to FIG. **5**. FIG. **8** is a plan view of semiconductor device **100** according to the third embodiment. FIG. **8** corresponds to FIG. **6**. FIG. **9** is a plan view of terminal **4** according to the third embodiment. FIG. **9** corresponds to FIG. **4**. Semiconductor device **100** shown in FIGS. **7** to **9** basically has the same configuration as that of semiconductor device **100** shown in FIGS. **5** and **6** and can achieve the same effect as that of semiconductor device **100** shown in FIGS. **5** and **6**. However, semiconductor device **100** shown in FIGS. **7** to **9** is different from semiconductor device **100** shown in FIGS. **5** and **6** in that notch portion h is provided at an end of terminal **4** in the width direction.

[0086] From a different point of view, notch portion h may be opened toward tip end surface **41***s***1** and one of the pair of side surfaces **41***s***2**.

[0087] With such a configuration, the width of signal terminal **41** in the y direction can be reduced. As a result, semiconductor device **100** can be reduced in size in the y direction. In addition, the material of signal terminal **41** can be reduced.

<Functions and Effects>

[0088] According to semiconductor device **100** described above, notch portion h is provided at the end of terminal **4** in the width direction.

[0089] With such a configuration, the width of signal terminal **41** in the y direction can be reduced. As a result, semiconductor device **100** can be reduced in size in the y direction. In addition, the material of signal terminal **41** can be reduced.

Fourth Embodiment

<Configuration of Semiconductor Device>

[0090] FIG. **10** is a side view of semiconductor device **100** according to a fourth embodiment. FIG. **10** corresponds to FIG. **5**. FIG. **11** is a partially enlarged side view of a region XI shown in FIG. **10**. FIG. **11** corresponds to FIG. **3**. Semiconductor device **100** shown in FIGS. **10** and **11** basically has the same configuration as that of semiconductor device **100** shown in FIGS. **5** and **6** and can achieve the same effect as that of semiconductor device **100** shown in FIGS. **5** and **6**. However, semiconductor device **100** shown in FIGS. **10** and **11** is different from semiconductor device **100** shown in FIGS. **5** and **6** in that a tip end of terminal **4** is inclined with respect to the z direction.

[0091] As terminal **4**, signal terminal **41** has an inclined portion **45**. Inclined portion **45** is disposed at tip end region **41***a* of signal terminal **41**. Inclined portion **45** is inclined to be located between connection portion **6***b* and vertex portion **6**T in the z direction, which is a direction perpendicular to main surface **11***s* of substrate **1**.

[0092] By inclining the tip end of signal terminal **41** in the z direction as described above, the position of notch portion h in the z direction can be changed. As a result, deformation of metal wire **6** during manufacturing of semiconductor device **100** due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin can be suppressed.

<Functions and Effects>

[0093] According to semiconductor device **100** described above, metal wire **6** has vertex portion **6**T and connection portion **6***b*. Vertex portion **6**T is farthest from semiconductor element **2**. Connection portion **6***b* is in contact with terminal **4**. Terminal **4** has inclined portion **45**. Inclined portion **45** is located between connection portion **6***b* and vertex portion **6**T in the direction perpendicular to main surface **11***s* of substrate **1**.

[0094] With such a configuration, contact between metal wire **6** and the other members, and contact between metal wires **6** adjacent to each other are suppressed.

[0095] As a result, the occurrence of a short-circuit failure in semiconductor device **100** is suppressed.

Fifth Embodiment

<Configuration of Semiconductor Device>

[0096] FIG. **12** is a plan view of signal terminal **41** according to a fifth embodiment. FIG. **12** corresponds to FIG. **4**. Semiconductor device **100** shown in FIG. **12** basically has the same configuration as that of semiconductor device **100** shown in FIGS. **1** to **4** and can achieve the same effect as that of semiconductor device **100** shown in FIGS. **1** to **4**. However, semiconductor device **100** shown in FIG. **12** is different from semiconductor device **100** shown in FIGS. **1** to **4** in that notch portion h has a pair of inclined surfaces **41***s***3**. Specifically, notch portion h has the pair of inclined surfaces **41***s***3**, an inner wall surface **41***s***4** and a bottom surface **41***s***5**. The pair of inclined surfaces **41***s***3**, inner wall surface **41***s***4** and bottom surface **41***s***5** form notch portion h.

[0097] Each of inclined surfaces **41***s***3** is continuous to side surface **41***s***2** and inner wall surface **41***s***4**. Bottom surface **41***s***5** is a surface of notch portion h that is most recessed from the tip end of signal terminal **41** in the x direction. Bottom surface **41***s***5** is continuous to inner wall surface **41***s***4**.

[0098] A portion of inclined surface **41***s***3** continuous to inner wall surface **41***s***4** is disposed between a portion of inclined surface **41***s***3** continuous to side surface **41***s***2** and bottom surface **41***s***5** in the x direction. Thus, each of the pair of inclined surfaces **41***s***3** is inclined toward a center A**1** of signal terminal **41** in the width direction. From a different point of view, as shown in FIG. **12**, in a plan view of main surface **11***s*, each of inclined surfaces **41***s***3** is inclined to face center A**1**. That is, a distance between the pair of inclined surfaces **41***s***3** becomes gradually shorter from the tip end of signal terminal **41** toward bottom surface **41***s***5**. Thus, it becomes easier to insert metal wire **6** into notch portion h.

<Functions and Effects>

[0099] According to semiconductor device **100** described above, notch portion h has inclined surface **41***s***3**. Inclined surface **41***s***3** is inclined toward center A**1**.

[0100] With such a configuration, it becomes easier to insert metal wire **6** into notch portion h.

Sixth Embodiment

<Configuration of Semiconductor Device>

[0101] FIG. **13** is a plan view of signal terminal **41** according to a sixth embodiment. FIG. **13** corresponds to FIG. **12**. Semiconductor device **100** shown in FIG. **13** basically has the same configuration as that of semiconductor device **100** shown in FIG. **12** and can achieve the same effect as that of semiconductor device **100** shown in FIG. **12**. However, semiconductor device **100** shown in FIG. **13** is different from semiconductor device **100** shown in FIG. **12** in that notch portion h has a return surface **41***s***6**. Specifically, return surface **41***s***6** protrudes from inner wall surface **41***s***4** toward center A**1** (see FIG. **12**). Return surface **41***s***6** is continuous to inner wall surface **41***s***4** and inclined surface **41***s***3**. Return surface **41***s***6** faces bottom surface **41***s***5**.

[0102] With such a configuration, push-out of metal wire **6** to the outside of notch portion h during manufacturing of semiconductor device **100** due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin is suppressed.

<Functions and Effects>

[0103] According to semiconductor device **100** described above, notch portion h has return surface **41***s***6**. Return surface **41***s***6** protrudes toward center A**1** (see FIG. **12**).

[0104] With such a configuration, push-out of metal wire **6** to the outside of notch portion h during manufacturing of semiconductor device **100** due to vibrations generated during transport of semiconductor device **100** or the injection pressure of the resin during injection of the resin is suppressed.

Seventh Embodiment

[0105] A power conversion device to which the semiconductor device described in any one of the above-described first to sixth embodiments is applied will now be described. Although the present disclosure is not limited to a particular power conversion device, application of the present disclosure to a three-phase inverter will be described below as a seventh embodiment.

[0106] FIG. **14** is a block diagram showing a configuration of a power conversion system to which the power conversion device according to the present embodiment is applied. The power conversion system shown in FIG. **14** is constituted of a power supply **400**, a power conversion device **200** and a load **300**. Power supply **400** is a DC power supply and supplies DC power to power conversion device **200**. Power supply **400** can be configured by a variety of types, and can be configured by a DC system, a solar battery or a storage battery, for example. Power supply **400** may be configured by a rectifier circuit or an AC/DC converter connected to an AC system. Alternatively, power supply **400** may be configured by a DC/DC converter that converts DC power output from the DC system into prescribed power.

[0107] Power conversion device **200** is a three-phase inverter connected between power supply **400** and load **300**, and converts DC power supplied from power supply **400** into AC power and supplies the AC power to load **300**. As shown in FIG. **14**, power conversion device **200** includes a main conversion circuit **201** that converts DC power into AC power and outputs the AC power, and a control circuit **203** that outputs, to main conversion circuit **201**, a control signal for controlling main conversion circuit **201**.

[0108] Load **300** is a three-phase electric motor driven by the AC power supplied from power conversion device **200**. Load **300** is not limited to a specific application, and load **300** is an electric motor mounted on various types of electric devices and is used as an electric motor for a hybrid vehicle, an electric vehicle, a railroad vehicle, an elevator, or an air-conditioning device, for example.

[0109] Details of power conversion device **200** will be described below. Main conversion circuit **201** includes a switching element and a freewheeling diode (not shown). When the switching element is switched, DC power supplied from power supply **400** is converted into AC power, which is supplied to load **300**. While there are various types of specific circuit configurations for main conversion circuit **201**, main conversion circuit **201** according to the present embodiment is a two-level three-phase full-bridge circuit and can be formed of six switching elements and six freewheeling diodes that are in antiparallel with the switching elements, respectively.

[0110] Semiconductor device **100** according to at least any one of the above-described first to sixth embodiments is applied as a semiconductor module **202** to at least any one of the switching elements and the freewheeling diodes of main conversion circuit **201**. The six switching elements have every two switching elements connected in series to form upper and lower arms, and the upper and lower arms configure the full bridge circuit's phases (a U phase, a V phase and a W phase). Output terminals of the upper and lower arms, i.e., three output terminals of main conversion circuit **201** are connected to load **300**.

[0111] Main conversion circuit **201** includes a drive circuit (not shown) that drives each switching element, and main conversion circuit **201** may have the drive circuit built into semiconductor module **202**, or may include the drive circuit separately from semiconductor module **202**. The drive circuit generates a drive signal for driving the switching elements of main conversion circuit **201**, and supplies the drive signal to a control electrode of each switching element of main conversion circuit **201**. Specifically, in accordance with the control signal from below-described control circuit **203**, a drive signal for bringing a switching element into an on state and a drive signal for bringing a switching element into an off state are output to the control electrode of each switching element. When the switching element is maintained in the on state, the drive signal is a voltage signal (ON signal) equal to or higher than a threshold voltage of the switching element. When the switching element is maintained in the off state, the drive signal is a voltage signal (OFF signal) equal to or lower than the threshold voltage of the switching element.

[0112] Control circuit **203** controls the switching elements of main conversion circuit **201** such that desired power is supplied to load **300**. Specifically, control circuit **203** calculates a time for which each switching element of main conversion circuit **201** should be turned on (ON time) based on the power to be supplied to load **300**. For example, control circuit **203** can control main conversion circuit **201** by PWM control by which an ON time of a switching element is modulated in accordance with a voltage to be output. Control circuit **203** outputs a control command (control signal) to the drive circuit of main conversion circuit **201** such that the ON signal is output to a switching element to be turned on at each point in time and the OFF signal is output to a switching element to be turned off at each point in time. In response to this control signal, the drive circuit outputs the ON signal or the OFF signal as the drive signal to the control electrode of each switching element.

[0113] In the power conversion device according to the present embodiment, semiconductor device **100** according to any one of the above-described first to sixth embodiments is applied as semiconductor module **202** to at least any one of the switching elements and the freewheeling diodes of main conversion circuit **201**. Therefore, the electrical insulating properties can be improved, and the reliability of the power conversion device can be improved.

[0114] Although the example in which the present disclosure is applied to a two-level three-phase inverter has been described in the present embodiment, the present disclosure is not limited thereto, and is applicable to various power conversion devices. Although a two-level power conversion device has been described in the present embodiment, a three-level power conversion device or a multi-level power conversion device may be adopted, and when electric power is supplied to a single-phase load, the present disclosure may be applied to a single-phase inverter. When electric power is supplied to a DC load or the like, the present disclosure is also applicable to a DC/DC converter or an AC/DC converter.

[0115] The power conversion device to which the present disclosure is applied is not limited to the above case where the load is an electric motor. For example, the power conversion device can also be used as a power supply device for an electric discharge machine, a laser beam machine, an induction heating cooking device, or a non-contact power feeding system, and furthermore can also be used as a power conditioner for a solar power generation system, a power storage system, or the like.

[0116] It should be noted that the semiconductor devices described in the embodiments can be variously combined as needed. Moreover, for the dependent claims in the scope of claims for

patent, dependent forms corresponding to the combinations are also intended to be included.
[0117] Although the embodiments of the present disclosure have been described, it should be understood that the embodiments disclosed herein are illustrative and non-restrictive in every respect. The scope of the present disclosure is defined by the terms of the claims and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

Claims

1. A semiconductor device comprising: a substrate; a semiconductor element disposed on the substrate; a terminal disposed at a position farther than the semiconductor element when viewed from the substrate; and a metal wire connecting the semiconductor element and the terminal, wherein a notch portion is provided in the terminal, and a part of the metal wire is inserted into the notch portion.
2. The semiconductor device according to claim 1, wherein the semiconductor element has a main electrode and a signal electrode, and the terminal has a main terminal connected to the main electrode with a joining portion interposed therebetween, and a signal terminal connected to the signal electrode via the metal wire.
3. The semiconductor device according to claim 1, wherein the terminal includes a tip end region overlapping with the substrate in a plan view of the substrate, and an external region other than the tip end region, and the metal wire is connected to the external region.
4. The semiconductor device according to claim 1, wherein the notch portion is provided at a center of the terminal in a width direction.
5. The semiconductor device according to claim 4, wherein the notch portion has an inclined surface inclined toward the center.
6. The semiconductor device according to claim 4, wherein the notch portion has a return surface protruding toward the center.
7. The semiconductor device according to claim 1, wherein the notch portion is provided at an end of the terminal in a width direction.
8. The semiconductor device according to claim 1, wherein the metal wire has a vertex portion farthest from the semiconductor element, and a connection portion that is in contact with the terminal, and the terminal has an inclined portion located between the connection portion and the vertex portion in a direction perpendicular to a main surface of the substrate.
9. A power conversion device comprising: a main conversion circuit having the semiconductor device according to claim 1, to convert input power and output the converted input power; and a control circuit to output a control signal for controlling the main conversion circuit to the main conversion circuit.